Revisiting Madry's Defense: An Independent Study on Model Capacity and Adversarial Robustness

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October 2025 - January 2026

Abstract

This study investigates the adversarial resilience of deep neural networks through a principled re-evaluation of the min-max optimization framework. Using a "clean-room" implementation[1], we assess the impact of representational capacity on robustness by contrasting compact CNN architectures with high-capacity WideResNet-28-10 models across CIFAR-10 and the German Traffic Sign Recognition Benchmark (GTSRB). Our results empirically validate that model capacity is a primary driver of robustness, as evidenced by the superior convergence stability of residual architectures under iterative L_∞ attacks. Furthermore, we demonstrate that stochastic training regimes effectively mitigate the inherent accuracy-robustness trade-off, acting as a critical regularizer for low-capacity models. Crucially, we extend our analysis to localized L_0 perturbations, revealing that L_∞ adversarial training induces a significant cross-norm transfer of robustness against universal adversarial patches. By validating the generalization of robust optimization against Momentum Iterative Method (MIM) and spatial adversaries, this work reinforces the practical viability of adversarial training in safety-critical computer vision domains.

[1]The source code is available at `https://github.com/Monthaonos/Advanced_machine_learning`.

1 Introduction

The last decade has witnessed the unprecedented success of Deep Convolutional Neural Networks (CNNs) in computer vision, achieving super-human performance on benchmarks ranging from ImageNet classification to autonomous navigation. However, this predictive power has revealed a startling vulnerability: the existence of *adversarial examples*. First identified by Szegedy et al. [9] and further analyzed by Goodfellow et al. [4], these are inputs modified by imperceptible perturbations that cause models to output incorrect predictions with high confidence.

This discovery triggered an "arms race" between attackers developing more sophisticated gradient-based perturbations and defenders proposing heuristics to block them. For a significant period, this race was dominated by the phenomenon of *gradient masking*: defenses that did not truly make models robust but merely broke the gradient flow, providing a false sense of security until stronger optimization attacks were designed to bypass them [1]. In this context, the work of Madry et al. [7] marked a pivotal paradigm shift. By reframing adversarial robustness not as a mere patch but as a formal optimization problem—specifically a min-max saddle point problem—they provided a principled framework for training robust models.

A central insight of Madry's study was the hypothesis that **robustness requires significantly larger model capacity** than standard classification. Intuitively, while standard models minimize loss by latching onto brittle "non-robust" features, robust models are forced to ignore these shortcuts and learn invariant geometric shapes. This study aims to verify this hypothesis through an independent "clean-room" re-implementation. We investigate the relationship between architecture size and robust performance across disparate distributions: organic images (CIFAR-10) and the rigid geometry of traffic signs (GTSRB). Furthermore, we extend the investigation to L_0 "patch" attacks to assess if L_∞-robust training provides cross-norm security.

Our Contributions. In this independent study, we provide a validation of the current gold standard in robust optimization. Our specific contributions are as follows:

1. **Capacity Analysis:** We empirically verify the necessity of model capacity by contrasting the failure of compact CNNs against the success of high-capacity WideResNet-28-10 architectures.
2. **Stochastic Regularization:** We demonstrate that for limited-capacity models, a *probabilistic training strategy* prevents model collapse and offers a tunable trade-off between natural accuracy and adversarial robustness.
3. **First-Order Universality:** We confirm that robustness against PGD generalizes to other first-order adversaries, specifically the Momentum Iterative Method (MIM).
4. **Safety-Critical Extension & Cross-Norm Robustness:** We evaluate these findings on the German Traffic Sign Recognition Benchmark (GTSRB) and show that L_∞ adversarial training yields significant, albeit incomplete, resilience against localized L_0 patch attacks.

1.1 Adversarial Threat Model: From Intuition to Min-Max

We consider a standard classification task where the goal is to predict labels y from input images x. The standard approach involves defining a loss function L (e.g., cross-entropy) and finding the parameters θ that minimize the expected risk on the data distribution $\mathcal{D}$:

$$\theta^* = \arg\min_\theta \mathbb{E}_{(x,y)\sim\mathcal{D}}[L(f_\theta(x), y)]$$

where $f_\theta(\cdot)$ denotes the classifier's prediction function.

Before formalizing the adversarial objective, it is crucial to understand the physical nature of an attack. Intuitively, it acts as an **optical illusion** for the model. We take a clean input x and superimpose a specific "noise" or perturbation δ to generate a new example $x_{adv} = x + \delta$. The core concept is that while the model correctly classifies the clean input (i.e., $f_\theta(x) = y$), the perturbation is maliciously engineered so that:

$$f_\theta(x_{adv}) \neq y \tag{1}$$

Crucially, while δ drastically changes the model's prediction, it must remain imperceptible to the human eye. Figure 1 illustrates this phenomenon on a GTSRB traffic sign.

Formally, we assume an adversary can add a perturbation δ provided it remains within a defined set S. Following Madry et al. [7], we define this set as an ℓ_∞-ball of radius ϵ:

$$S = \{\delta \in \mathbb{R}^d : \|\delta\|_\infty \leq \epsilon\}$$

The choice of the ℓ_∞ norm ensures that no single pixel is modified by more than ϵ, preserving the structural integrity of the image.

The goal of robust optimization is to train a network that performs well on the **worst-case** perturbations within S, yielding the min-max saddle point problem:

$$\min_\theta \rho(\theta) \quad \text{where} \quad \rho(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\max_{\delta \in S} L(f_\theta(x+\delta), y) \right] \tag{2}$$

1.2 First-Order Adversarial Attacks

To evaluate and train our models, we consider three primary first-order gradient attacks. While Madry et al. focused on FGSM and PGD, suggesting that PGD is the "universal" first-order adversary, we extend our evaluation to momentum-based methods to verify this hypothesis.

Fast Gradient Sign Method (FGSM). Introduced by Goodfellow et al. [4], FGSM is a one-step linearization of the inner maximization:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x L(f_\theta(x), y)) \tag{3}$$

While computationally efficient, FGSM relies on the assumption that the loss surface is linear around x. In practice, robust models often mask their gradients or have highly non-linear loss landscapes, rendering single-step attacks like FGSM insufficient for evaluation.

Iterative Attacks (PGD & MIM). Unlike single-step methods, iterative attacks refine the perturbation over T steps. We initialize $x^{(0)}$ randomly within the ϵ-ball, iterate the update rule, and define the final adversary as $x_{adv} = x^{(T)}$. This random initialization and iterative refinement are critical to explore the perturbation space and escape poor local maxima.

Projected Gradient Descent (PGD). The standard PGD update is:

$$x^{(t+1)} = \Pi_{x+S}\left(x^{(t)} + \alpha \cdot \text{sign}(\nabla_x L(f_\theta(x^{(t)}), y))\right) \tag{4}$$

Momentum Iterative Method (MIM). While PGD serves as the primary baseline in standard evaluations, we introduce MIM to challenge our models against a broader range of first-order attacks. MIM accumulates a velocity vector $g^{(t)}$ over iterations [3]:

$$g^{(t+1)} = \mu \cdot g^{(t)} + \frac{\nabla_x L}{\|\nabla_x L\|_1}; \quad x^{(t+1)} = \Pi_{x+S}(x^{(t)} + \alpha \cdot \text{sign}(g^{(t+1)})) \tag{5}$$

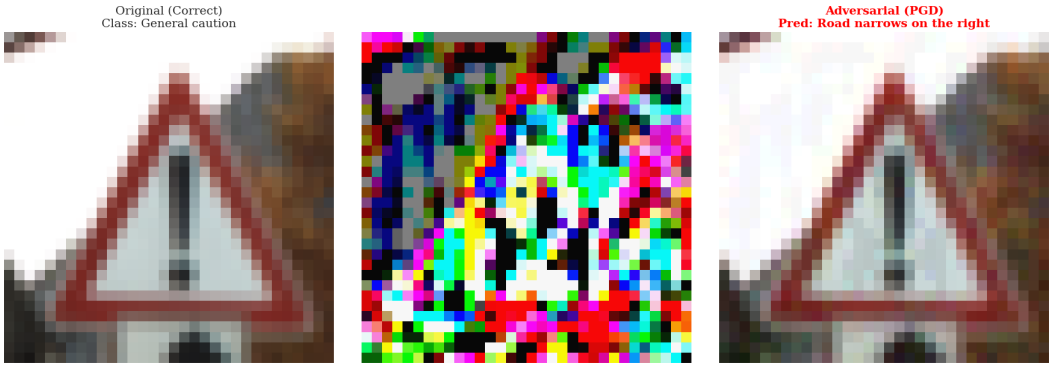


Figure 1: Visualizing an adversarial attack. The original image x (left) is modified by an amplified perturbation δ (center) to create x_{adv} (right). Under a constraint $\epsilon = 8/255$, the robust model is deceived into predicting an incorrect class.

By smoothing the gradient directions, MIM stabilizes the update trajectory and helps escape poor local maxima, often yielding adversarial examples that are more transferable and effective against robust models than standard PGD.

Universal Adversarial Patches (L_0 Attacks). Unlike L_∞ perturbations, patch attacks simulate localized but intense L_0 noise by overlaying a learnable pattern onto a specific region of the image [2]. The adversarial image x_{adv} is formally constructed as:

$$x_{adv} = (1 - M) \odot x + M \odot P \tag{6}$$

In this formulation, x represents the clean input, $M \in \{0, 1\}$ is a binary mask defining the occlusion area, and P is the adversarial pattern, both of the same dimensions as the input image. The symbol $\odot$ denotes the Hadamard product, which enables the precise spatial blending of the sticker with the original pixels. To ensure the patch is *universal*, we solve $P^* = \arg\max_P \mathbb{E}_{(x,y)\sim\mathcal{D}}[L(f_\theta(x_{adv}), y)]$ using Gradient Ascent. Our implementation stabilizes this optimization through three key mechanisms: (i) **spatial invariance**, where P is randomly repositioned within a central region (25% to 75% of image dimensions) to target semantic features rather than background noise; (ii) **gradient accumulation**, averaging updates over 4 batches to smooth the optimization trajectory and filter out batch-specific noise; and (iii) **box constraints**, where the Adam optimizer's [6] updates are projected via a clamping operation to maintain $P \in [0, 1]$.

We optimize the patch P over a subset of the training set to create a *universal* sticker. This attack is specifically evaluated on GTSRB to mirror real-world threats where an adversary might physically deface a traffic sign.

1.3 Adversarial Training Defense Strategy

The core of our defense strategy relies on directly optimizing Equation 2. Since the inner maximization problem has no closed-form solution for deep networks, we approximate it using the strongest attack available during training.

We adopt the **Adversarial Training** protocol. Instead of training on the clean dataset $\mathcal{D}$, we construct an adversarial dataset $\hat{\mathcal{D}}$ on the fly. The training algorithm proceeds as follows for each mini-batch $\mathcal{B} = \{(x_i, y_i)\}$:

1. **Attack Generation:** For each sample x_i, we find an approximate worst-case perturbation δ_i^* using K-step PGD.
2. **Parameter Update:** We update the network parameters θ by minimizing the adversarial loss over the mini-batch $\mathcal{B}$:

$$\theta \leftarrow \theta - \eta \cdot \nabla_\theta \frac{1}{|\mathcal{B}|} \sum_i L(f_\theta(x_i + \delta_i^*), y_i)$$

This process effectively populates the "blind spots" of the model, forcing it to learn feature representations that are invariant to small ℓ_∞ variations.

1.4 Network Architectures and Capacity

A central hypothesis of this work is that robust optimization requires significantly larger model capacity than standard training. To test this, we evaluate three distinct architectures:

- **Small CNN:** A shallow network consisting of four convolutional layers (32 and 64 filters) followed by fully connected layers. While this model achieves $> 85\%$ accuracy on standard CIFAR-10, it serves as a proxy for limited representational capacity. Its shallow depth restricts the complexity of the features it can extract.
- **ResNet-18:** To bridge the gap between the shallow CNN and the massive WideResNet, we incorporate the standard ResNet-18 architecture introduced by He et al. [5]. Characterized by its residual skip connections, it enables the training of deeper networks without gradient degradation. We deploy this architecture primarily for the GTSRB dataset to investigate if an intermediate capacity is sufficient for tasks with simpler geometric features.

- **WideResNet-28-10:** Following Zagoruyko and Komodakis [11], we implement a Wide Residual Network. Unlike standard ResNets which increase depth, WideResNets increase the width (number of channels) by a factor $k = 10$. This width is crucial for adversarial robustness: it allows the network to learn a larger variety of feature detectors in parallel, necessary to capture the subtle, non-robust features.

Our rationale was to select architectures with sufficiently disparate capacities to empirically verify if model size is indeed a necessary condition for robustness.

1.5 Mitigating Capacity Constraints via Probabilistic Training

Standard adversarial training replaces 100% of natural examples with their adversarial counterparts. For limited-capacity models, we observed that this leads to "adversarial overfitting," where the model fails to converge on meaningful boundaries.

To address this, we propose a *Batch-wise Probabilistic Training* strategy. We introduce a Bernoulli random variable $B \sim \text{Bernoulli}(P_{train})$ at each training step. The effective loss function switches between standard and adversarial modes:

$$\mathcal{L}_{batch}(\theta, \mathcal{B}) = \begin{cases} \frac{1}{|\mathcal{B}|} \sum L(f_\theta(\text{PGD}(x)), y) & \text{if } B = 1 \text{ (prob. } P_{train}) \\ \frac{1}{|\mathcal{B}|} \sum L(f_\theta(x), y) & \text{if } B = 0 \text{ (prob. } 1 - P_{train}) \end{cases} \tag{7}$$

This approach acts as a regularizer. By periodically exposing the model to clean data, we prevent the weights from diverging into a regime where they only recognize specific adversarial patterns but lose the general semantics of the image.

1.6 Evaluation Metrics and Objectives

We evaluate the performance of our models using two primary metrics: *Natural Accuracy* ($\mathcal{A}_{nat}$) on clean data and *Robust Accuracy* ($\mathcal{A}_{rob}$) on adversarial examples generated via Projected Gradient Descent (PGD).

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ be the dataset, where x_i is the input image and y_i the class label. Let f_θ denote the classifier with parameters θ.

Accuracy Definitions. Natural accuracy measures the classification performance on unmodified test samples:

$$\mathcal{A}_{nat} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(f_\theta(x_i) = y_i) \tag{8}$$

where $\mathbb{1}(\cdot)$ is the indicator function. Conversely, robust accuracy measures the model's stability under the worst-case perturbation δ within a defined feasible set $\mathcal{S}$ (typically an ℓ_p-norm ball, $\|\delta\|_p \leq \epsilon$):

$$\mathcal{A}_{rob} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(f_\theta(x_i + \delta_i^*) = y_i) \tag{9}$$

where $\delta_i^* = \arg\max_{\delta \in \mathcal{S}} \mathcal{L}(f_\theta(x_i + \delta), y_i)$ is the adversarial perturbation that maximizes the loss. A model is considered truly robust only if $\mathcal{A}_{rob}$ remains high. However, it is well-documented that there exists an inherent trade-off [10]: increasing robustness often degrades natural accuracy.

Loss Functions and Optimization. To train these models, we minimize specific loss functions. The *Standard Loss* ($\mathcal{L}_{nat}$) minimizes the empirical risk using the Cross-Entropy (CE) loss ℓ:

$$\mathcal{L}_{nat}(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(f_\theta(x), y)] \tag{10}$$

For adversarial training, we solve the min-max saddle point problem to minimize the *Robust Loss* ($\mathcal{L}_{rob}$):

$$\mathcal{L}_{rob}(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\max_{\delta \in \mathcal{S}} \ell(f_\theta(x + \delta), y) \right] \tag{11}$$

Here, the inner maximization is typically approximated using multi-step PGD during training.

Attack Success Rate (ASR). While robust accuracy evaluates the model's remaining performance, the Attack Success Rate (ASR) specifically quantifies the effectiveness of the adversarial patch. Unlike $\mathcal{A}_{rob}$, ASR is calculated exclusively on the subset of samples that the model correctly classifies in their natural state.

Let $\mathcal{I}_{corr} = \{i \in \{1, \ldots, N\} : f_\theta(x_i) = y_i\}$ be the set of indices corresponding to initially correctly classified images. The ASR is defined as the proportion of these samples for which the adversarial patch P successfully induces a misclassification:

$$ASR = \frac{1}{|\mathcal{I}_{corr}|} \sum_{i \in \mathcal{I}_{corr}} \mathbb{1}(f_\theta(x_{adv,i}) \neq y_i) \tag{12}$$

where $x_{adv,i}$ is the patched image generated according to Eq. 6. This metric is particularly relevant for universal attacks, as it isolates the deception power of the learned pattern P from the model's inherent generalization errors on clean data.

2 Experimental Framework and Results

2.1 Datasets

To evaluate our models across different complexity levels and domains, we employ two distinct benchmarks.

CIFAR-10. Serving as the primary benchmark for adversarial robustness research, CIFAR-10 consists of 60,000 32×32 color images distributed across 10 classes (50,000 training, 10,000 testing). Its low resolution combined with high intra-class variability makes it a challenging testbed, requiring significant model capacity to learn robust features.

GTSRB (German Traffic Sign Recognition Benchmark). To assess robustness in a safety-critical application, we extend our study to the GTSRB dataset [8]. It comprises 43 classes of traffic signs with approximately 39,200 training and 12,600 test images.

2.2 Experimental Setup

We conduct our experiments on the CIFAR-10 and GTSRB datasets using the PyTorch framework. Our setup is designed to evaluate robustness across varying model capacities and attack modalities.

Optimization Strategy. We employ distinct optimization protocols tailored to the architecture's complexity to ensure convergence and prevent robust overfitting:

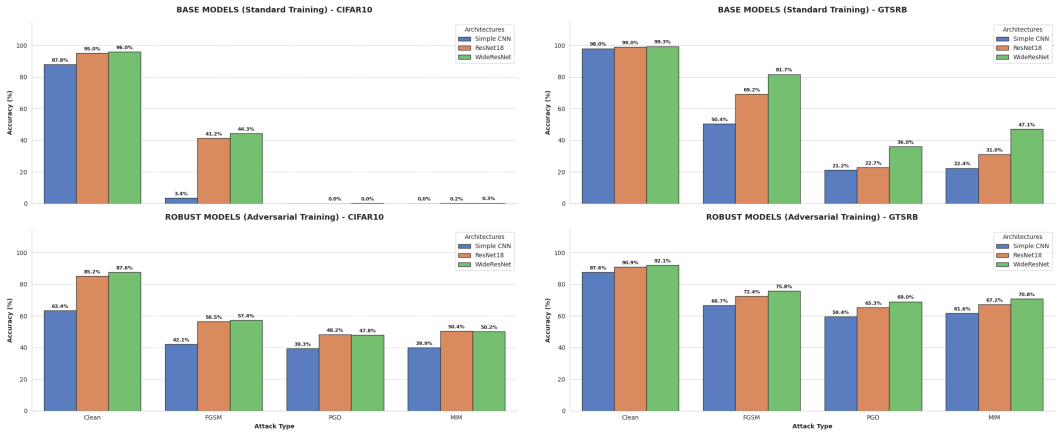
- **Small CNN:** Trained for 30 epochs using the Adam optimizer [6] with a learning rate of 10^{-3}. This reduced epoch count and adaptive optimizer choice are specifically selected to mitigate early overfitting observed in low-capacity models.
- **ResNet-18 & WideResNet:** These high-capacity models are trained for 100 epochs using Stochastic Gradient Descent (SGD) with a momentum of 0.9 and a weight decay of 5×10^{-4}. We initialize the learning rate at 0.1 and employ a *Cosine Annealing* scheduler. This approach facilitates smoother convergence towards the global minimum without the need for manual decay milestone tuning.

Adversarial Training Parameters (L_∞). For robust model variants, we follow the standard benchmark settings: a perturbation budget $\epsilon = 8/255$ and a PGD step size $\alpha = 2/255$. The inner maximization is solved using 6 iterations (PGD-6) during the training phase to balance computational efficiency and adversarial strength.

Universal Patch Setup (L_0). The localized patch attack is optimized using the Adam optimizer for 1000 iterations. The patch is configured with a scale factor of 0.25, effectively covering 25% of the input image surface. We use a fixed learning rate of 0.05 for the patch pixels to ensure rapid yet stable convergence of the adversarial pattern P.

2.3 Impact of Model Capacity on Robustness

Our experiments empirically validate the central hypothesis of Madry et al. [7]: model capacity is a critical prerequisite for adversarial robustness. This relationship is illustrated through a comparative analysis across three architectures of increasing complexity: *Simple CNN*, *ResNet-18*, and *WideResNet-28-10*.



(a) CIFAR-10 Capacity Analysis

(b) GTSRB Capacity Analysis

Figure 2: Comparative performance of Base (Standard Training) vs. Robust (Adversarial Training) models across architectures. Accuracy metrics are reported for Clean, FGSM, PGD, and MIM attacks.

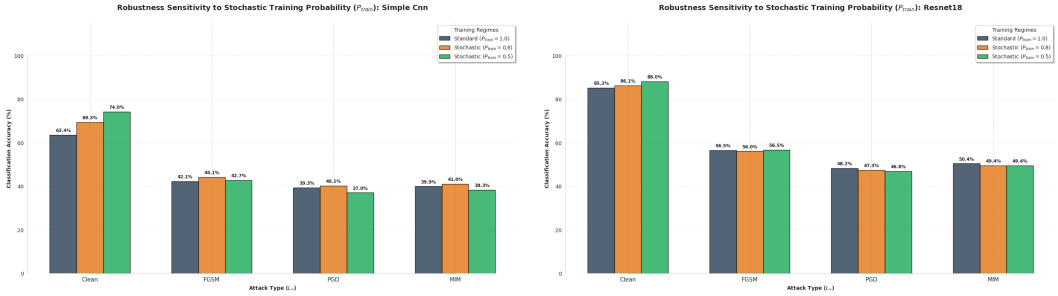
As shown in Figures 2a and 2b, we observe a clear hierarchy of robustness driven by representational capacity:

- **Capacity as a Robustness Driver:** In both datasets, the transition from a shallow CNN to residual architectures yields substantial gains in robust accuracy. While standard models across all capacities collapse to nearly 0% accuracy under iterative PGD attacks on CIFAR-10, adversarial training enables higher-capacity models to retain significant performance (e.g., $\sim$48.2% for ResNet-18 and $\sim$47.8% for WideResNet). This confirms that wider and deeper networks are better equipped to learn the complex decision boundaries required to ignore brittle, non-robust features and focus on global geometric structures. The convergence stability of this min-max optimization is further evidenced by the training and adversarial loss trajectories provided in Appendix A.
- **Dataset Complexity and Geometric Simplicity:** A striking observation is that all models perform significantly better on GTSRB than on CIFAR-10. For instance, the robust WideResNet achieves 69.0% PGD accuracy on GTSRB versus 47.8% on CIFAR-10. We attribute this to the nature of the data: traffic signs consist of high-contrast, simple geometric shapes (circles, triangles) that are easier for the model to internalize as invariant features. Conversely, CIFAR-10's organic and textured images provide more "non-robust" shortcuts that the adversary can exploit.
- **Natural Robustness in GTSRB:** Notably, the base WideResNet on GTSRB exhibits a "natural" robustness of 36.0% against PGD even without adversarial training. This suggests that high-capacity models, when trained on simple geometric data, naturally develop filters that are partially resistant to first-order perturbations, whereas they remain completely vulnerable on CIFAR-10.

2.4 Evaluation of Stochastic Training Regimes

Pure adversarial training ($P_{train} = 1.0$), while highly effective at maximizing robustness, typically imposes a significant "accuracy tax" on clean data (natural accuracy). To mitigate this phenomenon, we evaluated a stochastic training strategy consisting of injecting a variable proportion of clean samples into adversarial mini-batches. This approach is controlled by the parameter P_{train}, representing

the probability that a given sample is converted into an adversarial example via PGD during the training phase.



(a) Simple CNN Sensitivity

(b) ResNet18 Sensitivity

Figure 3: Impact of the stochastic training probability (P_{train}) on natural and robust accuracy for Simple CNN and ResNet18 architectures.

The analysis of Figures 3a and 3b reveals several critical insights regarding the accuracy-robustness trade-off:

- **Accuracy-Robustness Trade-off:** For both architectures, decreasing P_{train} leads to a systematic increase in natural accuracy. In the case of ResNet18, shifting from $P_{train} = 1.0$ to $P_{train} = 0.5$ yields a gain of nearly 3 percentage points in clean accuracy (from 85.2% to 88.0%), while conceding only a marginal loss in PGD robustness (from 48.2% to 46.8%).
- **Regularization for Low-Capacity Models:** The benefit is even more pronounced for the Simple CNN. Employing a moderate stochastic regime ($P_{train} = 0.8$) offers the optimal balance, actually surpassing pure adversarial training in PGD robustness (40.1% vs. 39.3%) while simultaneously improving natural accuracy by 6 points (69.3% vs. 63.4%). This suggests that for capacity-limited models, exposure to clean data acts as a necessary regularizer to stabilize decision boundaries.
- **Stability Against First-Order Attacks:** Notably, robustness against MIM attacks remains stable or improves slightly at $P_{train} = 0.8$ for the Simple CNN (41.0% vs. 39.9%). This observation validates the hypothesis that stochastic training helps prevent "adversarial overfitting," where the model becomes overly specialized to the training attack at the expense of global generalization.

In conclusion, stochastic training provides a critical control mechanism. It allows for the dynamic adjustment of the equilibrium between security and utility, proving particularly essential for maintaining the practical viability of compact models in environments where computational constraints limit architectural capacity.

2.5 Synthesis of Universal Patch Attacks (L_0)

To conclude our study, we evaluate the vulnerability of our models to a localized L_0 attack. Unlike the PGD attack which introduces imperceptible noise globally, the Universal Patch represents a concentrated perturbation (covering 25% of the image) that remains identical across the entire test set. We report the *Attack Success Rate* (ASR), representing the percentage of correctly classified images that were successfully fooled by the patch.

The results in Table 1 highlight several key findings:

- **High Vulnerability of Standard Models:** Standard models are extremely susceptible to the patch, with ASR exceeding 90% on GTSRB for the Simple CNN. This confirms that without robust optimization, localized high-frequency patterns easily hijack the model's decision-making.
- **Cross-Norm Robustness Transfer:** Although the adversarial training was conducted using L_∞ PGD, it provides a significant "transfer" of robustness to L_0 attacks. For CIFAR-10, the ASR for the Simple CNN drops drastically from 89.7% to 14.8%.

Table 1: Comparative Synthesis of Universal Patch Attack Success Rates (ASR %) for Standard (Clean) vs. Adversarially Trained (Robust) models.

	CIFAR-10		GTSRB	
Architecture	**Clean ASR**	**Robust ASR**	**Clean ASR**	**Robust ASR**
Simple CNN	89.7%	14.8%	93.6%	55.6%
ResNet-18	71.3%	27.1%	64.8%	51.4%
WideResNet	67.3%	31.0%	71.9%	53.1%

- **Dataset Specificity:** Robust models on GTSRB remain more vulnerable to patches ($\sim$51–55% ASR) compared to CIFAR-10. This suggests that while the simple geometry of traffic signs aids L_∞ robustness, it also makes the models more sensitive to localized occlusions that cover key sign features.

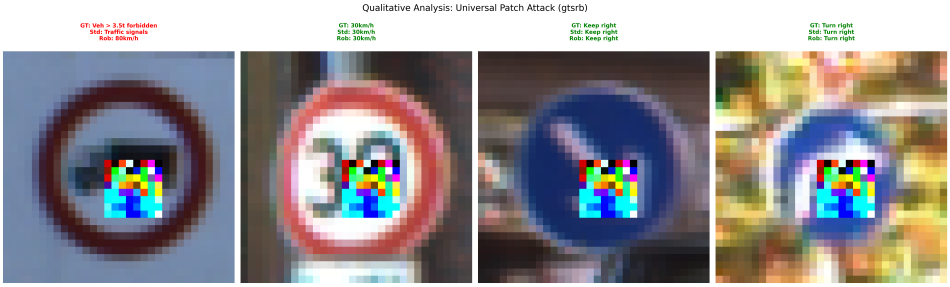


Figure 4: Qualitative visualization of the universal L_0 patch attack on the GTSRB dataset using the ResNet-18 architecture.

3 Conclusion

In this study, we revisited the foundational claims of Madry et al. regarding the relationship between model capacity and adversarial robustness. Through an independent implementation and evaluation across multiple architectures and datasets, we confirmed that model capacity is indeed a primary driver of robustness. Our results demonstrate that while shallow networks struggle to converge under full adversarial constraints, high-capacity architectures like WideResNet-28-10 successfully internalize robust features, achieving nearly 48% robust accuracy on CIFAR-10 and up to 69% on GTSRB.

Our investigation yielded three key insights beyond the original PGD framework. First, we demonstrated that the "accuracy tax" of adversarial training can be mitigated through stochastic training regimes (P_{train}). This is particularly vital for compact models, where a probabilistic exposure to clean data acts as a necessary regularizer to maintain utility without sacrificing security. Second, our extension to the GTSRB dataset highlighted the influence of data geometry; the inherent simplicity of traffic signs allows models to achieve higher robust baselines compared to the textured complexity of CIFAR-10. Finally, our analysis of universal L_0 patch attacks revealed a significant cross-norm transfer effect: models trained against global L_∞ perturbations develop a structural "shape-bias" that enables them to partially resist localized, intense occlusions.

In conclusion, while the "arms race" in adversarial machine learning continues, the min-max optimization framework remains the most reliable pathway toward trustworthy vision systems. Our findings suggest that for safety-critical applications like traffic sign recognition, combining high-capacity backbones with stochastic training strategies offers a viable balance between performance and resilience. Future work should focus on formalizing these cross-norm defenses to ensure security against heterogeneous attack vectors in real-world environments.

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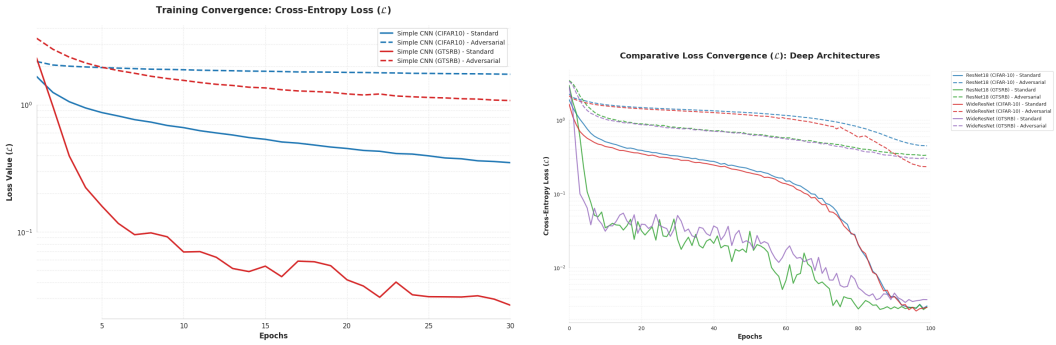
A Training Dynamics and Loss Trajectories

This appendix provides the loss evolution during the training of our robust models, **supporting the capacity analysis presented in Section 2.3**. These curves illustrate the stability of the min-max optimization across different architectures and datasets.



(a) CIFAR-10 Loss (WideResNet)

(b) GTSRB Loss (ResNet-18)

Figure 5: Evolution of the Cross-Entropy loss during adversarial training. The convergence of the adversarial loss (inner maximization) confirms the stability of the PGD-6 training protocol.